Quiz 2

● Graded

Student

Colin Skinner

Total Points

31 / 35 pts

Question 1

Instructions

0 / 0 pts

+ 0 pts Incorrect

✔ **+ 0 pts** Correct

Question 2

(no title)

3 / 5 pts

– 0 pts Correct

– 0 pts **Rounding:** We awarded full points since your answer is correct, but note for next time to round to the specified number of decimal places.

✔ **– 1 pt** **Missing Step:** Forgot to apply Bayes' rule to obtain $p(\text{sufficient} \mid \text{sensor})$

✔ **– 1 pt** **Missing step:** Forgot to compute $p(\text{sensor reading “sufficient/insufficient”})$

– 1 pt **Incorrect Calculation:** Mistake in calculating term(s) in the VOI formula

– 4 pts **Partial Credit:** Made attempt of calculating VOI (but went down incorrect route)

– 5 pts Incorrect

Question 3

(no title)

5 / 5 pts

✔ **+ 5 pts** Correct

+ 4.5 pts Minor issue

+ 3.5 pts Almost correct

+ 2 pts Incorrect answer, work shown

+ 0 pts Incorrect

Question 4

(no title) **5** / 5 pts

✔ **+ 5 pts** Correct

+ 4 pts Incorrect distances/kernel values calculations.

+ 4 pts Incorrect kernel value calculations.

+ 0 pts Insufficient progress or no attempt.

+ 2 pts Major errors in set-up/kernel values calculation.

+ 4 pts Did not normalize kernel values or incorrect normalization.

+ 3 pts Missing 1 of 3 kernel values entirely.

Question 5

(no title) **5** / 5 pts

✔ **+ 5 pts** Correct

+ 2.5 pts Did not answer question 2

+ 3.25 pts One value calculated incorrectly

+ 2.5 pts Both values calculated incorrectly

+ 0 pts No answer / incorrect

Question 6

(no title) **5** / 5 pts

✔ **– 0 pts** Correct (Mentions Elite/Best Performing or High Utility Samples)

– 2 pts Does not mention how CEM improves performance/incorrect reasoning

– 5 pts Does not mention anything about how CEM improves performance or how it skips gradient computation

Question 7

(no title) **3** / 5 pts

– 0 pts Correct

✔ **– 2 pts** Added the gradient estimates for Boost and Glide instead of just computing the Boost gradient.

– 1 pt Forgot to multiply by $U(\tau)$ at the end.

– 1 pt Outputted matrix of Boost and Glide gradients instead of just Glide.

– 1 pt Made minor arithmetic or computation error (wrong e exponent, wrong sigmoid expression, etc.)

– 1 pt Wrong expression for log-likelihood

– 4 pts Partial attempt

– 5 pts Wrong answer and no work shown

So when the drone took the action :boost, the log-likelihood would only contain terms about the likelihood of what actions were taken at the recorded states.

Question 8

(no title) **5** / 5 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect normalization of the gradient step

– 1.5 pts Incorrect usage of step factor / wrong formulation

– 1.5 pts Calculation error

– 5 pts Did not attempt

Q1 Instructions
0 Points

You may consult any material (e.g., books, calculators, computer programs, and online resources), but you may not consult other people inside or outside of the class or large language models like *ChatGPT*.

The quiz is designed to be completed in 90 minutes, but we will grant you 120 minutes total to complete and submit your quiz (including uploading any images, handling any logistical issues, etc.) The timing on Gradescope is a hard cutoff.

Ed will not allow any public posts during that time. Correct answers will receive full credit, and partial credit may be awarded based on work shown. Hence, we encourage you to show your work. You can upload **either** a file **or** give a response in the free response field (LaTeX is possible using *two* $ signs before and after your math expression – so a total of four $ signs).

Out of fairness to all students, only material submitted during the allowed time will be graded.

As a reminder, all questions are equally weighted. Some questions will take longer than others. Feel free to answer the questions in any order you like.

Write your name in the box below.

Colin Skinner

The Galactic Robotics Olympics. You are part of the robot team *Bayesians*, competing across planets in high-stakes decision-making events. Think strategically, and help your team bring home the gold!

Q2
5 Points

You are piloting your team's spacecraft preparing to depart from Mars. Before takeoff, you must decide whether to launch immediately or refuel first.

- If you launch immediately, there is a 0.6 probability that the fuel is sufficient, in which case you receive a utility of $+12$; otherwise, the mission fails and your utility is -60.
- If you refuel before launching, success is guaranteed, and your utility is $+5$. (You may refuel even if the tank was already sufficient).

Before deciding, you may use an advanced sensor that reports whether the fuel is sufficient or not sufficient. The sensor correctly identifies the true fuel status with probability 0.95. Compute the value of information for the sensor.

Provide your answer rounded to two decimal places, such as 0.01 or 0.25. If the actual value is 1.025646, you should enter 1.03.

Prob 2.jpeg Download

Not knowing

$EU(a_i \mid o_s) = P(s_s \mid a_i, o_s)\, U(s_s) + P(s_N \mid a_i, o_s)\, U(s_N)$

$= (0.6)(12) + (0.4)(-60) = -16.8$

from waiting

$EU^*(o_s) = 5$ w/o sensor

knowing for sure

$EU(a_i \mid o_s) = P(s_s \mid a_i, o_s)\, U(s_s) + P(s_N \mid a_i, o_s)\, U(s_N)$

$= (0.95)(12) + (0.05)(-60) = 8.4$

$EU(a_w \mid o_s) = P(s_s \mid a_w, o_s)\, U(s_s) + P(s_N \mid a_w, o_s)\, U(s_N)$

$= (1)(5) + (0)(5) = 5$

$EU^*(o_s) = 8.4$ w/ sensor

for $o_N \rightarrow EU^*(o_N) = 5$ waiting

No observation so

$$VOI(O' \mid o) = \left(\sum_{o'} P(o' \mid o) EU^*(o, o')\right) - EU^*(o)$$

$VOI(O) = \left(\sum_{o'} P(o')\, EU^*(o')\right) - EU^*(\phi)$

$= P(o_s)\, EU^*(o_s) + P(o_N)\, EU^*(o_N) - EU^*(\phi)$

both w/ sensor

$= (0.65)(8.4) + (0.35)(5) - 5 = 8.40$

will be 5 regardless of if sensor is right or wrong

not observing anything, optimal action is to stay for

Q3
5 Points

In the next event of the Galactic Robotics Olympics, your autonomous drone competes in a planetary delivery challenge. The environment state can be either Calm (s_c) or Turbulent (s_t). At each decision point, the drone must choose whether to Fly (a_f) or Hover (a_h). The rewards are $R(s_c, a_f) = +5$, $R(s_c, a_h) = +2$, $R(s_t, a_f) = -1$, and $R(s_t, a_h) = +1$.

- When the drone flies in a calm environment, it stays calm with probability 0.6.
- When the drone hovers in a calm environment, it stays calm with probability 0.8.
- When the drone flies in a turbulent environment, it stays turbulent with probability 0.7.
- When the drone hovers in a turbulent environment, it stays turbulent with probability 0.5.

With a discount factor of $\gamma = 0.9$ and initial values $U_0(s_c) = U_0(s_t) = 5$, perform one Bellman backup and report the updated values $U_1(s_c)$ and $U_1(s_t)$.

Prob 3.png Download

$U_1(s_c, a_f) = R(s_c, a_f) + \gamma\left[T(s_c|s_c, a_f)V_0(s_c) + T(s_t|s_c, a_f)U(s_t)\right]$

$= 5 + (0.9)\left[(0.6)(5) + (0.4)(5)\right] = 9.5$ max

$V_1(s_c, a_h) = R(s_c, a_h) + \gamma\left[T(s_c|s_c, a_h)V_0(s_c) + T(s_t|s_c, a_h)U(s_t)\right]$

$= 2 + (0.9)\left[(0.8)(5) + (0.2)(5)\right] = 6.5$ ✗

$U_1(s_c) = 9.5$

$$U_{k+1}(s) = \max_a \left(R(s,a) + \gamma \sum_{s'} T(s' \mid s,a) U_k(s')\right)$$

$U_2(s_t, a_f) = R(s_t, a_f) + (0.9)\left[T(s_c|s_t, a_f)U(s_c) + T(s_t|s_t, a_f)U(s_t)\right]$

$= -1 + 0.9\left[(0.3)(5) + (0.7)(5)\right] = 3.5$ ✗

$U_2(s_t, a_h) = R(s_t, a_h) + (0.9)\left[T(s_c|s_t, a_h)U(s_c) + T(s_t|s_t, a_h)U(s_t)\right]$

$= 1 + 0.9\left[(0.5)(5) + (0.5)(5)\right] = 5.5$ max

$U_1(s_t) = 5.5$

Q4
5 Points

In the asteroid mapping event, your mining drone is learning to estimate the potential mineral yield at new locations. The measured utilities at three known sites (represented as two-dimensional coordinates) are $U([0,0]) = 1.5$, $U([1,0]) = 4.0$, and $U([0,2]) = 3.0$. Estimate the utility at $[0.6, 1.2]$ using a Gaussian kernel smoother with $\sigma = 1$ and the Euclidean distance metric.

Provide your answer rounded to two decimal places, such as 0.01 or 0.25. If the actual value is 1.025646, you should enter 1.03.

Prob 4.png Download

$U_\theta(s) = \theta^T \beta(s)$ $\quad \theta = [1.5, 4, 3]$

$\beta_i(s) = \frac{k(s, s_i)}{\sum k(s_i, s_i)}$

$k\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0.6\\1.2\end{bmatrix}\right) = \exp\left(-\frac{0.6^2 + 1.2^2}{2(1)}\right) = 0.4066$

$k\left(\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0.6\\1.2\end{bmatrix}\right) = \exp\left(-\frac{0.4^2 + 1.2^2}{2}\right) = 0.4493$ → Sum = 1.4624

$k\left(\begin{bmatrix}0\\2\end{bmatrix}, \begin{bmatrix}0.6\\1.2\end{bmatrix}\right) = \exp\left(-\frac{0.6^2 + 0.8^2}{2}\right) = 0.6065$

$\vec{\beta} = \left\{\frac{0.4066}{\text{sum}}, \frac{0.4493}{\text{sum}}, \frac{0.6065}{\text{sum}}\right\} = [0.278, 0.307, 0.415]$

$U_\theta([0.6, 1.2]) = \theta^T \beta = [1.5, 4, 3] \cdot [0.278, 0.307, 0.415]$

$= 0.417 + 1.228 + 1.244$

$U_\theta([0.6, 1.2]) = 2.89$

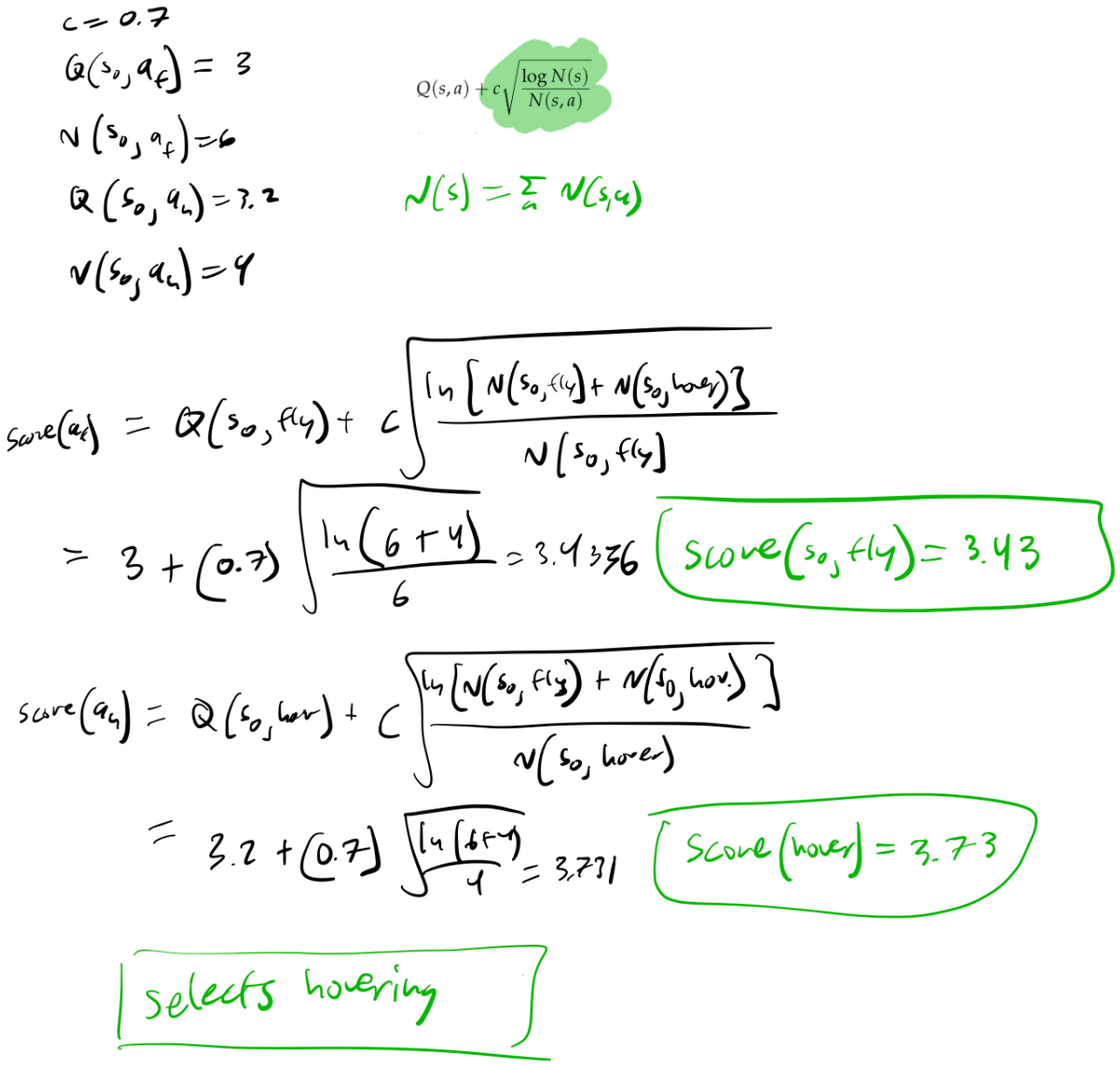
Q5
5 Points

Congrats! Your team (the *Bayesians*) has qualified for the planetary search-and-rescue round of competition, where your autonomous drone must quickly choose between two maneuvers: **Fly** and **Hover**. The team uses Monte Carlo Tree Search (MCTS) to plan in real time. At the current decision state s_0, the algorithm has accumulated the following statistics from past simulations: $Q(s_0, \text{Fly}) = 3.0$, $N(s_0, \text{Fly}) = 6$, $Q(s_0, \text{Hover}) = 3.2$, and $N(s_0, \text{Hover}) = 4$.
Using the UCB1 exploration heuristic with an exploration parameter of 0.7:

1. Compute the resulting scores for each action, **and**
2. Determine which action MCTS will select next from s_0.

Use the natural log ($\ln$) for any relevant calculations. Provide your answers rounded to two decimal places, such as 0.01 or 0.25. If the actual value is 1.025646, you should enter 1.03.

Prob 5.png Download



Q6

5 Points

The *Bayesians* now advance to the swarm evolution event, where coordinated drones evolve their flight policies over generations. You suggest using the cross-entropy method (CEM) to your teammates. In 1–3 sentences, explain how CEM is able to improve performance over time without relying on explicit gradients.

CEM simply takes a random distribution of points (typically a Gaussian), saving the time of calculating the gradients and then propagating the current samples in the directions of the graidient (which takes more time to compute). That is, simply focusing on the "elite samples" from this distribution yields the same result as calculating and climbing a gradient.

No files uploaded

Q7
5 Points

The arena falls silent as your drone faces its penultimate challenge: the flight-stability test. The drone has a stochastic policy $\pi_{\boldsymbol{\theta}}(a \mid s)$, parameterized by $\boldsymbol{\theta} = [\theta_1, \theta_2]$, chooses between two actions—Boost or Glide—using a logistic model with two parameters:

$$\pi_{\boldsymbol{\theta}}(\text{Boost} \mid s) = \frac{1}{1 + e^{-(\theta_1 s + \theta_2)}}, \quad \pi_{\boldsymbol{\theta}}(\text{Glide} \mid s) = 1 - \pi_{\boldsymbol{\theta}}(\text{Boost} \mid s).$$

You are given that for this policy,

$$\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\text{Boost} \mid s) = \big(1 - \pi_{\boldsymbol{\theta}}(\text{Boost} \mid s)\big) \begin{bmatrix} s \\ 1 \end{bmatrix},$$

$$\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\text{Glide} \mid s) = -\pi_{\boldsymbol{\theta}}(\text{Boost} \mid s) \begin{bmatrix} s \\ 1 \end{bmatrix}.$$

At state $s = 2$, the drone takes action Boost and receives total utility $U(\tau) = 6$. Compute the likelihood-ratio policy-gradient estimate: $\nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}) \approx U(\tau)\, \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid s)$ at $\boldsymbol{\theta} = [1, 0]$. (Since explicit gradient formulas are provided, you may use the single-trajectory estimate).

The values in your final answer should be rounded to two decimal places, such as 0.01 or 0.25. If the actual value is 1.025646, you should enter 1.03.

Prob 7.png Download

s_2, boost, $U(\tau) = 6$ $\quad \theta = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\nabla_\theta U(\theta) \approx U(\tau) \nabla_\theta \log \pi_\theta(a|s)$$

$$\pi_\theta(\text{Boost} \mid s^2) = \frac{1}{1 + e^{-(\theta_1 s + \theta_2)}} = \frac{1}{1 + e^{-(1.2 + 0)}} = 0.8808$$

$$\nabla_\theta \log \pi_\theta(\text{Boost}|s^2) = \begin{bmatrix} -\pi_\theta(\text{Boost}|s^2) \end{bmatrix} \begin{bmatrix} s^2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.8808 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.2384 \\ 0.1192 \end{bmatrix}$$

$$\nabla_\theta \log \pi_\theta(\text{Glide}|s') = -\pi_\theta(\text{Boost}|s) \begin{bmatrix} s \\ 1 \end{bmatrix}$$

$$= -0.8808 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1.7616 \\ -0.8808 \end{bmatrix}$$

$$\nabla_\theta U(\theta) \approx U(\tau) \left[\nabla_\theta \log \pi_\theta(\text{fly}|s^2) + \nabla_\theta \log \pi_\theta(\text{land}|s^2) \right]$$

$$\approx (6) \left(\begin{bmatrix} 0.2384 \\ 0.1192 \end{bmatrix} + \begin{bmatrix} -1.7616 \\ -0.8808 \end{bmatrix} \right) = \begin{bmatrix} -9.1391 \\ -4.570 \end{bmatrix}$$

$$\nabla_\theta U(\theta) \approx \begin{bmatrix} -9.14 \\ -4.57 \end{bmatrix}$$

Q8
5 Points

In the final round of the Galactic Robotics Olympics, the propulsion tuning event focuses on optimizing a robot's policy parameters $\boldsymbol{\theta} = [2, -1]$. The estimated gradient at this point is $\nabla U(\boldsymbol{\theta}) = [6, 8]$. The team performs gradient ascent to maximize $U(\boldsymbol{\theta})$. To prevent unstable thrust adjustments, we want to apply gradient scaling with an L_2 magnitude limit of 1 prior to taking a gradient step with step factor $\alpha = 0.5$. What is the updated parameter vector $\boldsymbol{\theta}'$ after one step?

Prob 8.png Download

$\theta = [2, -1]$ $\nabla U(\theta) = [6, 8]$ $\hat{\nabla U}(\theta) = \frac{[6,8]}{\sqrt{6^2+8^2}} = [0.6, 0.8]$

$\alpha = 0.5$ $\sqrt{2\varepsilon} = 1$

$\theta' = \theta + \alpha \nabla U(\theta) = [2, -1] + (0.5)[0.6, 0.8]$

$\theta' = [2.3, -0.6]$